

Meterswitch

Contributors: Vivien, Roberto, Michael, Noah

Main idea: Investigating intention by switching between two imagined meters in an ambiguous stimulus

Research Questions:

1. What is the timing of meter switching? Is there a transformation period that could demonstrate **intention**?
2. Does the switch timing differ between externally timed and internally timed switching?
3. Is the switching influenced by subject-level variables (musical experience) and stimulus-level variables (direction of switch)?

Experimental Setup: Behavioral Tapping Study, Computational Modeling

Meterswitch - Tapping Experiment

1. Priming



Or



2. Ambiguous rhythm



Switching:

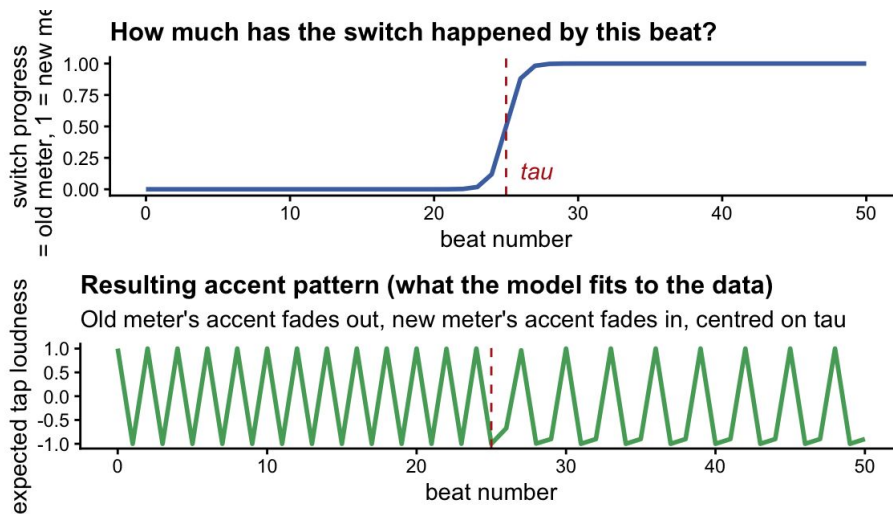
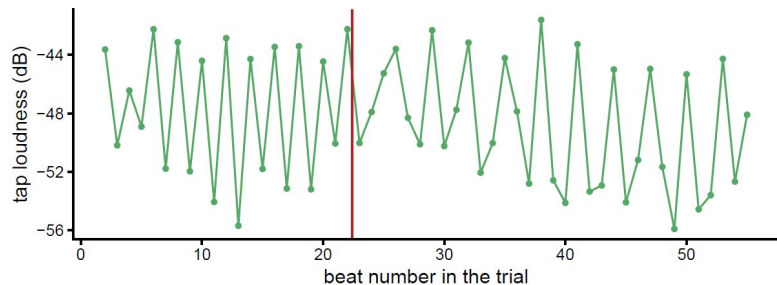
Cued

Spontaneous



Meterswitch - Pilot Results (N = 1)

1. Switching can be observed behaviorally and modelled computationally



Timing of the switch

There is a lag (~2.3 beats) between the cue and the estimated switch.

Switch speed might depend on musical training.

For spontaneous condition, the estimated switch time might provide a timestamp for EEG analysis.

Intention?

